MODULE – 3

DATA PROTECTION – RAID

Module 3: Data Protection – RAID

Upon completion of this module, you should be able to:

- Describe RAID implementation methods
- Describe the three RAID techniques
- Describe commonly used RAID levels
- Describe the impact of RAID on performance
- Compare RAID levels based on their cost, performance, and protection

Module 3: Data Protection – RAID

Lesson 1: RAID Overview

During this lesson the following topics are covered:

- RAID Implementation methods
- RAID array components
- RAID techniques

Why RAID?

RAID

It is a technique that combines multiple disk drives into a logical unit (RAID set) and provides protection, performance, or both.

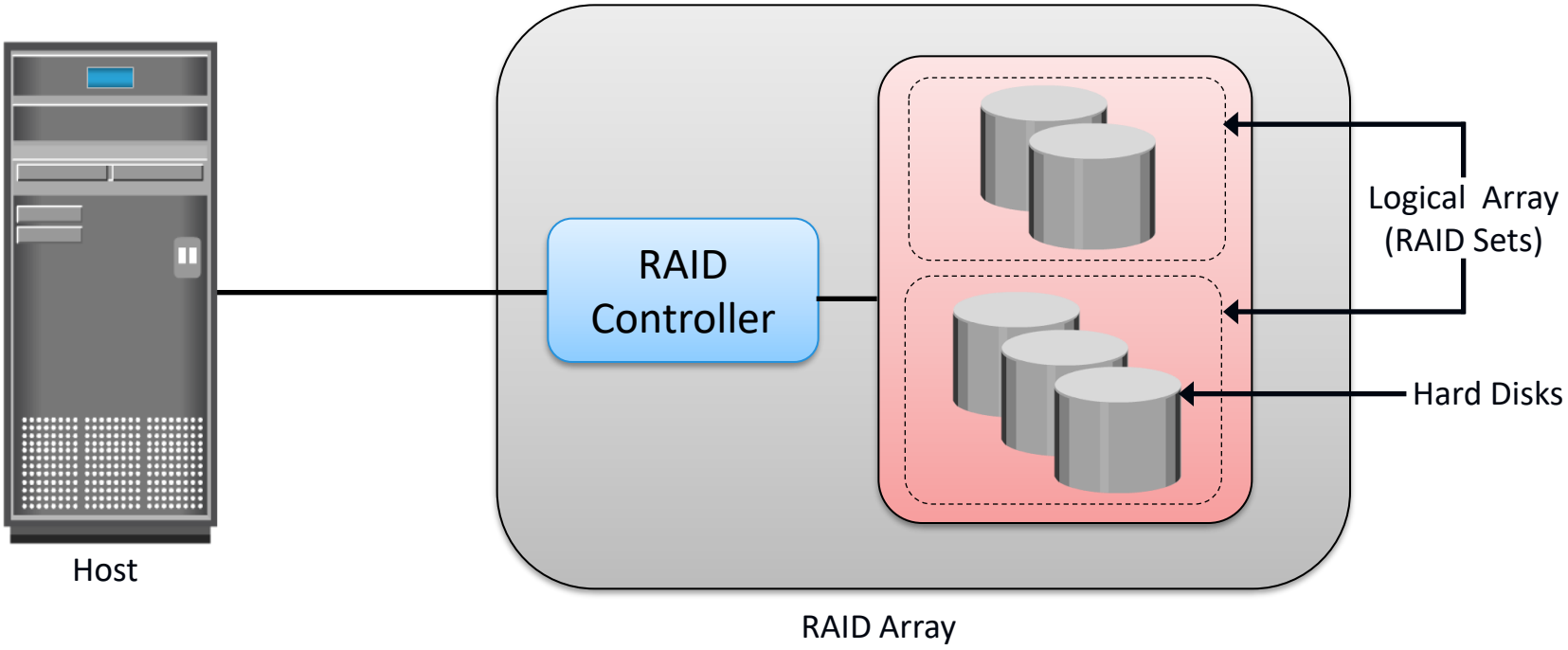
- Due to mechanical components in a disk drive it offers limited performance
- An individual drive has a certain life expectancy and is measured in MTBF (*Mean Time Between Failures*):
  - For example: If the MTBF of a drive is 750,000 hours, and there are 1000 drives in the array, then the MTBF of the array is 750 hours (750,000/1000)
- RAID was introduced to mitigate these problems

RAID Implementation Methods

- Software RAID implementation
  - Uses host-based software to provide RAID functionality
  - Limitations
    - Use host CPU cycles to perform RAID calculations, hence impact overall system performance
    - Support limited RAID levels
    - RAID software and OS can be upgraded only if they are compatible

- Hardware RAID Implementation
  - Uses a specialized hardware controller installed either on a host or on an array

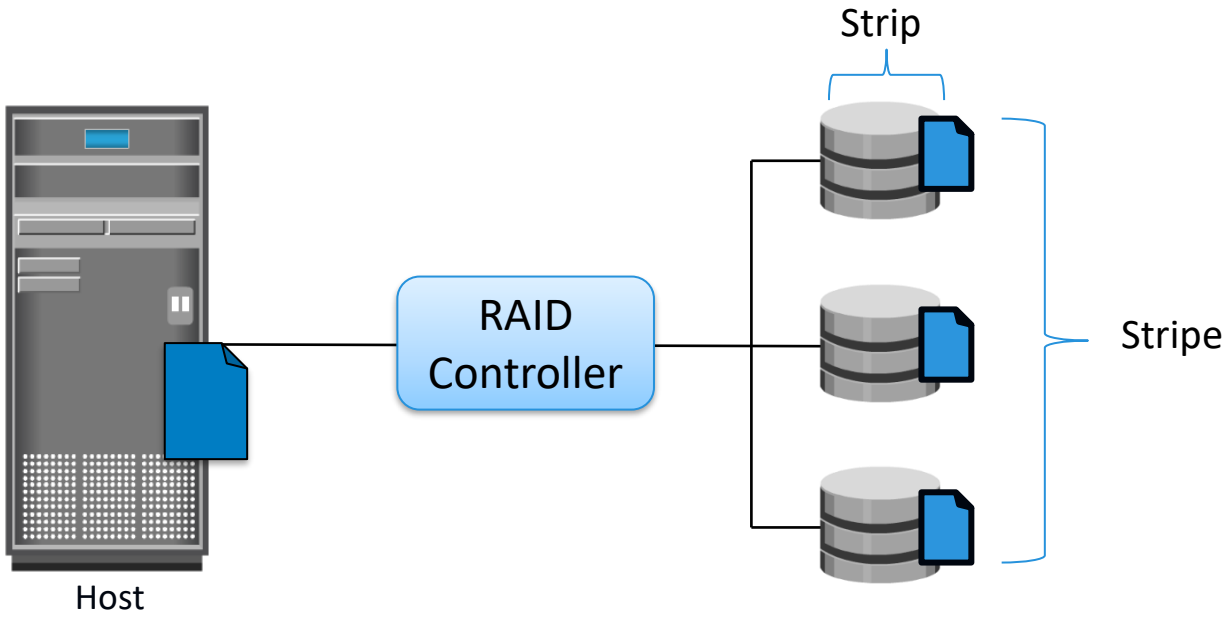
RAID Array Components



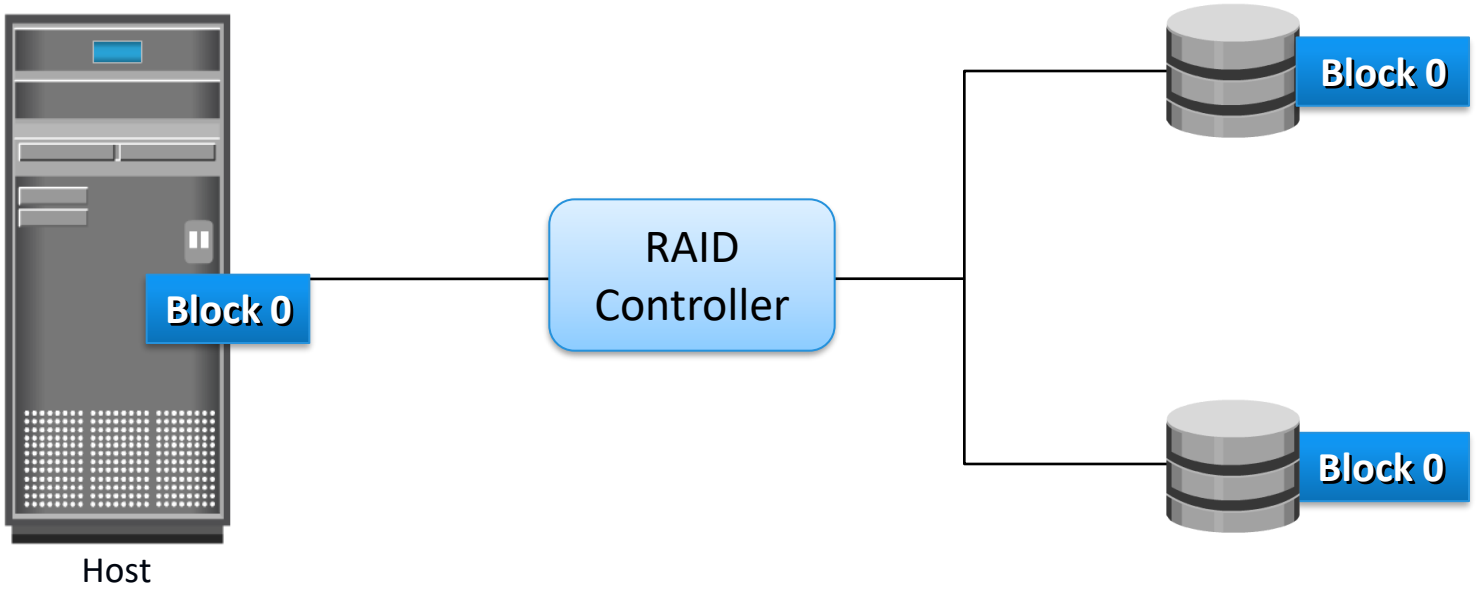
RAID Techniques

- **Striping**
  - Striping is a technique which offers the best performance of any RAID configuration. In a striped array, data is interleaved across all the drives in the array.
- **Mirroring**
  - whatever you write to one drive, gets written simultaneously to another. Thus, you always have an exact duplicate of your data on the second drive.
- **Parity**
  - RAID controller adds a parity (extra) byte of data tacked onto the actual data being written to the array. Added by the RAID controller to equal either an even or an odd number.
    - By analysing this value, the controller can determine whether the information has been compromised in any way. If it has, it can replace the data automatically with data from the other drive.

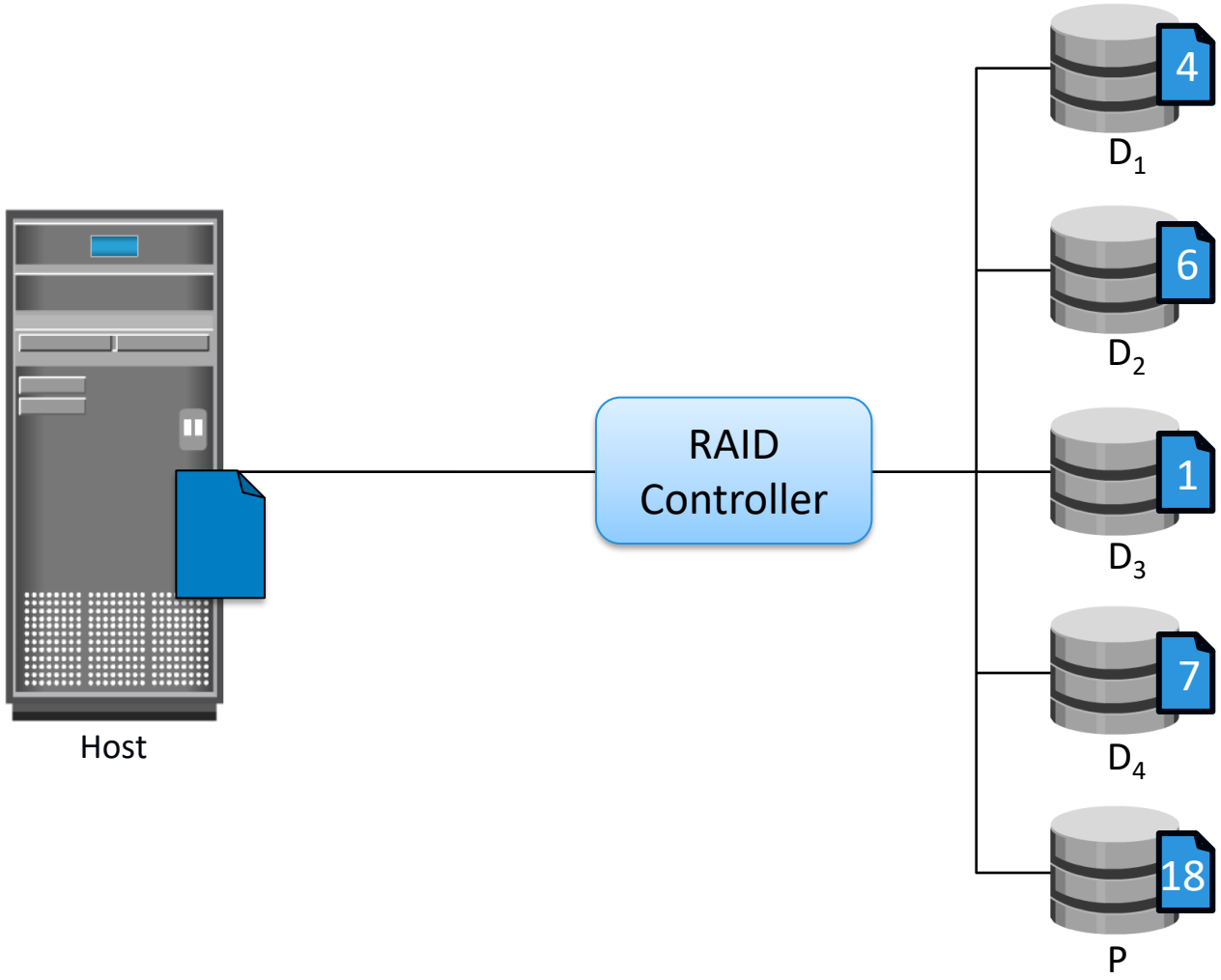
RAID Technique – Striping



RAID Technique – Mirroring

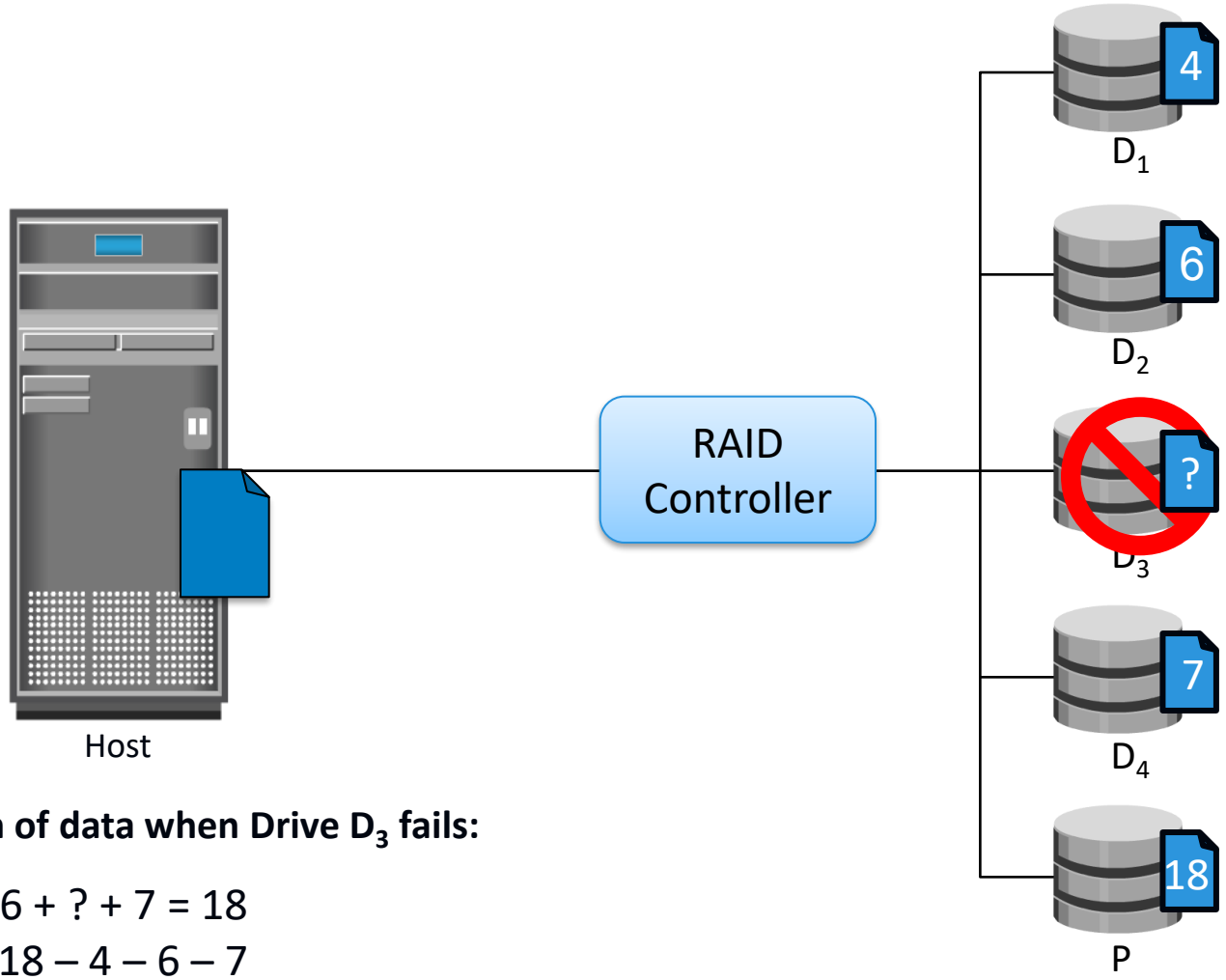


RAID Technique – Parity



Actual parity calculation is a bitwise XOR operation

Data Recovery in Parity Technique



Regeneration of data when Drive D_3 fails:

4 + 6 + ? + 7 = 18

? = 18 – 4 – 6 – 7

? = 1

Module 3: Data Protection – RAID

Lesson 2: RAID Levels

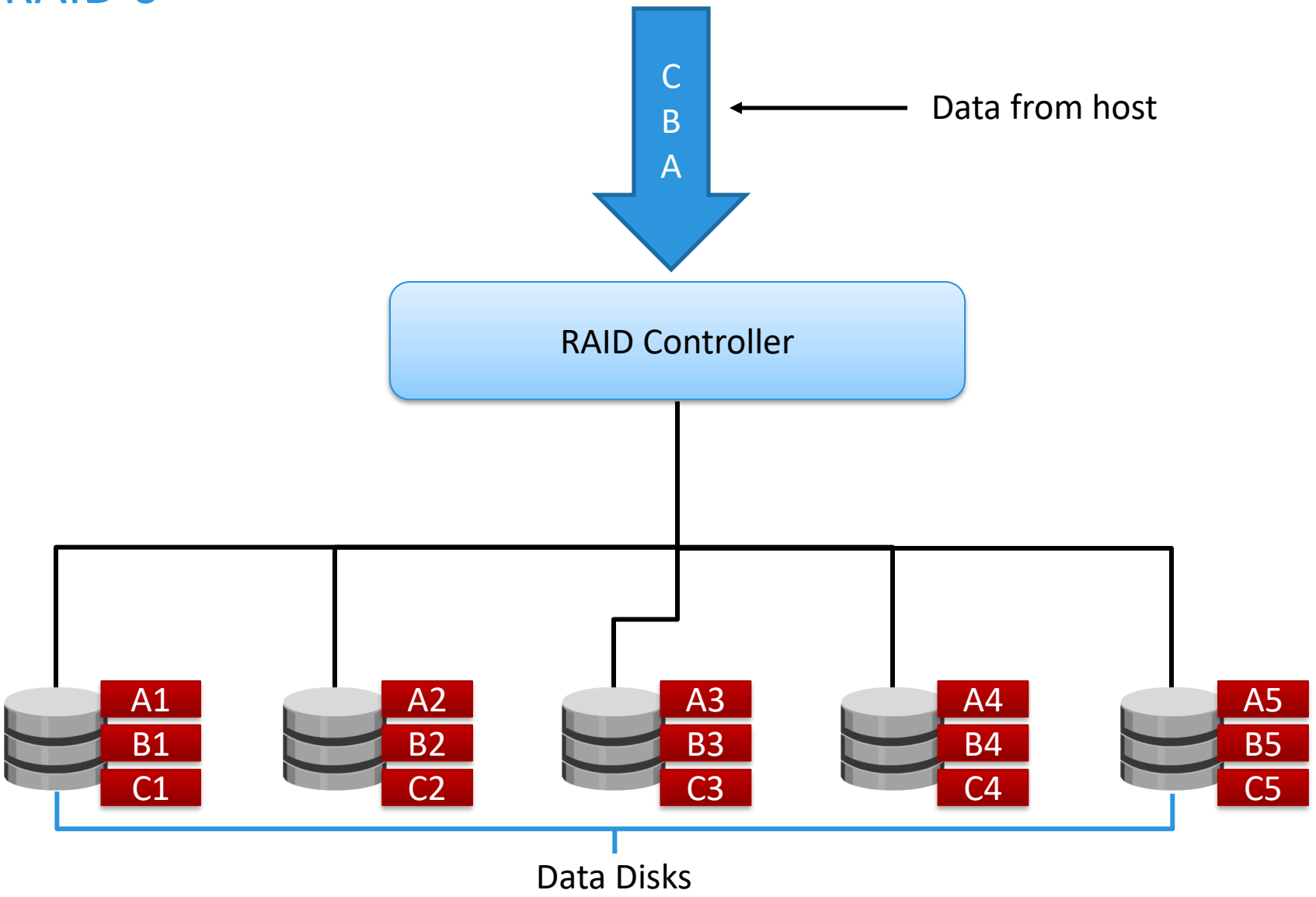
During this lesson the following topics are covered:

- Commonly used RAID levels
- RAID impacts on performance
- RAID comparison
- Hot spare

RAID Levels

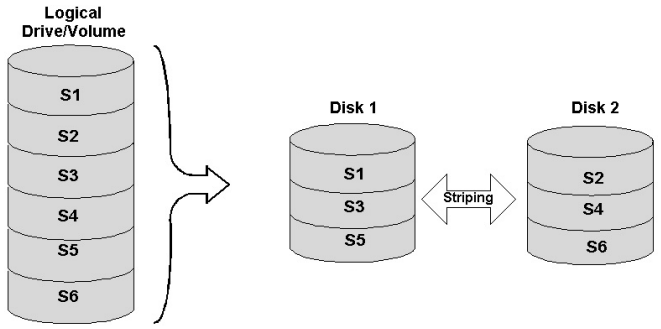
- Commonly used RAID levels are:
  - RAID 0 – Striped set with no fault tolerance
  - RAID 1 – Disk mirroring
  - RAID 1 + 0 – Nested RAID
  - RAID 3 – Striped set with parallel access and dedicated parity disk
  - RAID 5 – Striped set with independent disk access and a distributed parity
  - RAID 6 – Striped set with independent disk access and dual distributed parity

RAID 0

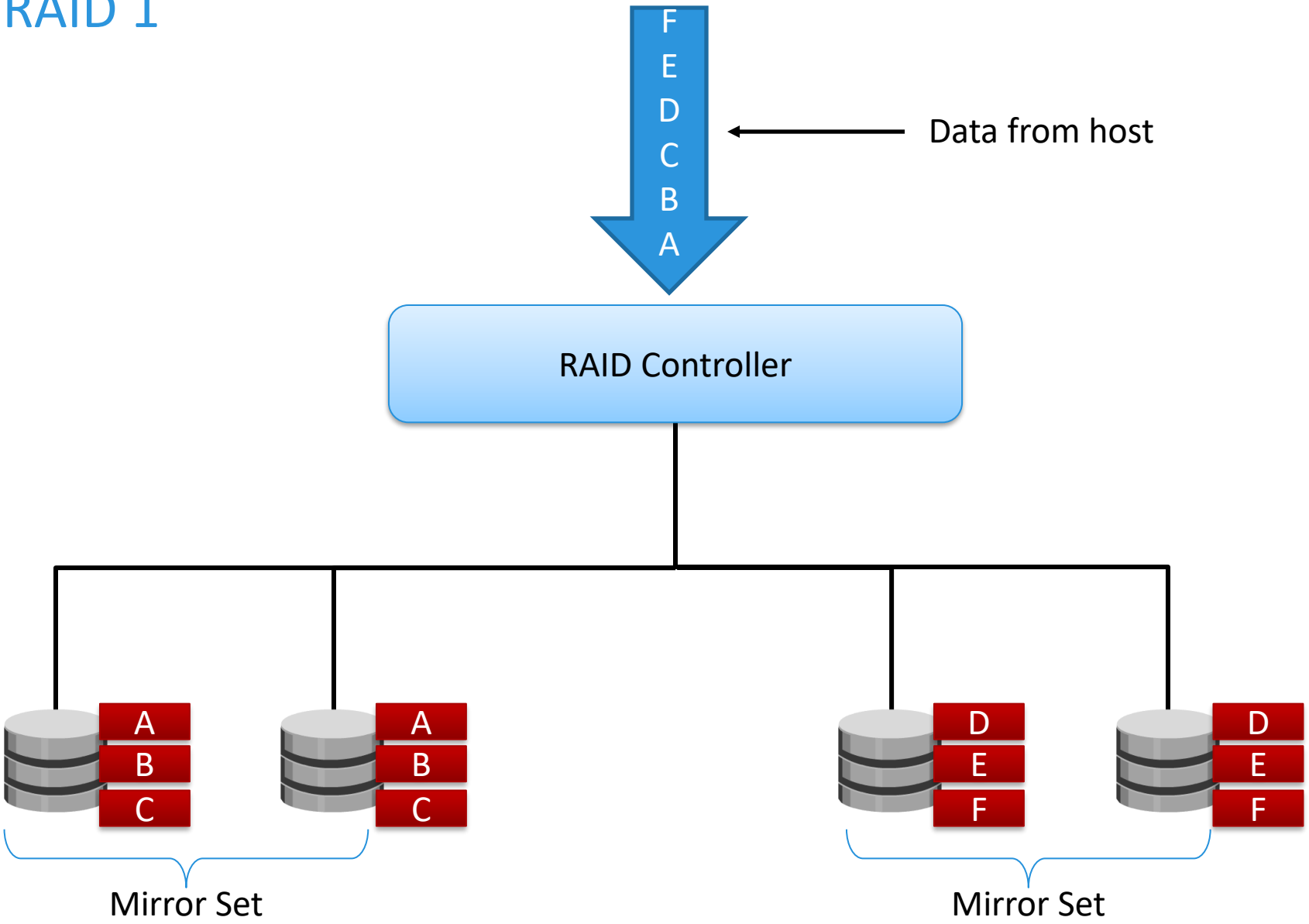

C
B
A
Data from host
RAID Controller
A1
B1
C1
A2
B2
C2
A3
B3
C3
A4
B4
C4
A5
B5
C5
Data Disks

RAID levels

- RAID 0 : non-redundant array of disk drives
- Also called "stripe" mode.
- Operations on the array will be split across the devices
  - e.g. a large write could be split up as 4 kB to disk 0, 4 kB to disk 1, 4 kB to disk 2, then 4 kB to disk 0 again, and so on.
- Pro: parallel writing increase speed nX where n = no. of partitions, x speed on single drive
- Con: No Redundancy, inefficient use of disk space: no savings!
- **RAID 0 provides no redundancy, and as such, should never be used for applications where data is critical**.

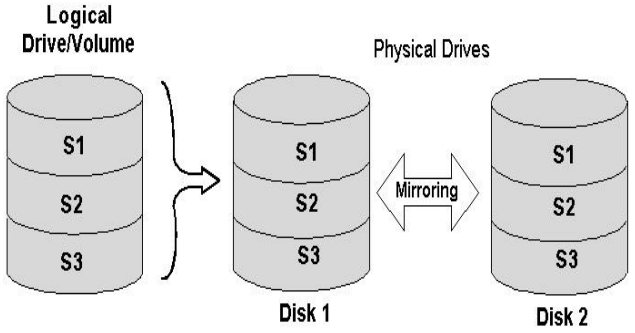


RAID 1

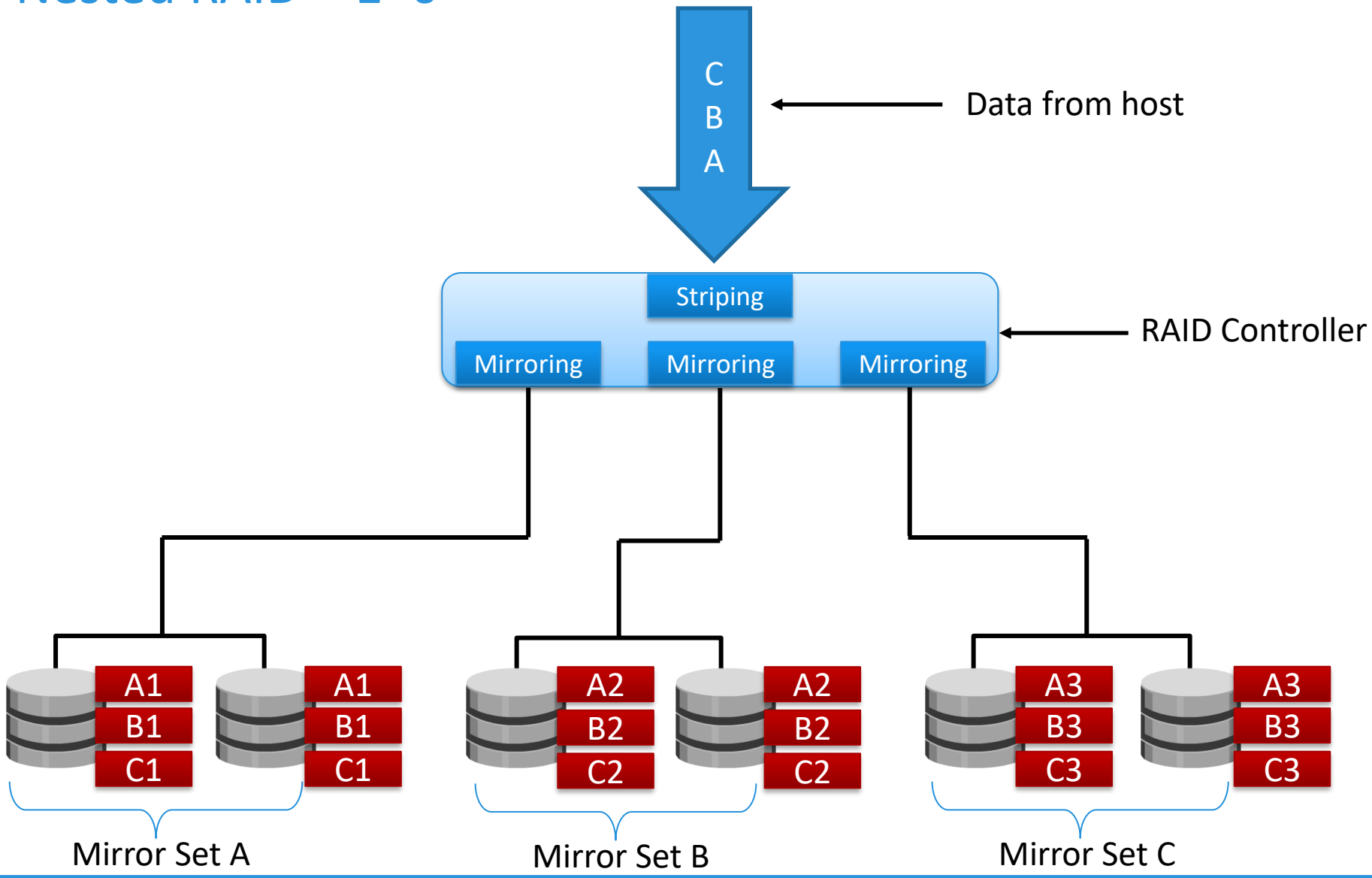

F
E
D
C
B
A
Data from host
RAID Controller
A
B
C
A
B
C
D
E
F
D
E
F
Mirror Set
Mirror Set

RAID 1

- **RAID-1** (Mirroring)
  - used on two or more disks.
  - This mode maintains an exact mirror of the information on one disk on the other disk(s).
- Pros: if disk removed (or crashes), all data is still intact
- Cons: Increase of cost proportional to the amount of redundancy.
- Performance may be impacted if copy is done serially.



Nested RAID – 1+0



RAID 3 & 4

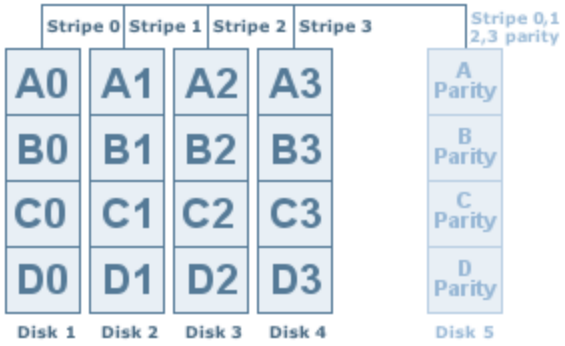
RAID 3

- Similar to RAID 2 which uses **byte level** striping and a dedicated parity disk
- BUT RAID 3 uses simple (&fast) XOR algorithm to generate parity.
- Data is striped (at the byte level) across multiple disks. The parity information is sent to a dedicated parity disk, but the failure of any disk in the array can be tolerated.
  - Pros: delivers good performance and fault tolerance.
  - Cons: dedicated parity disk does slow down write speeds because the parity info has to be written to the parity drive whenever a write occurs

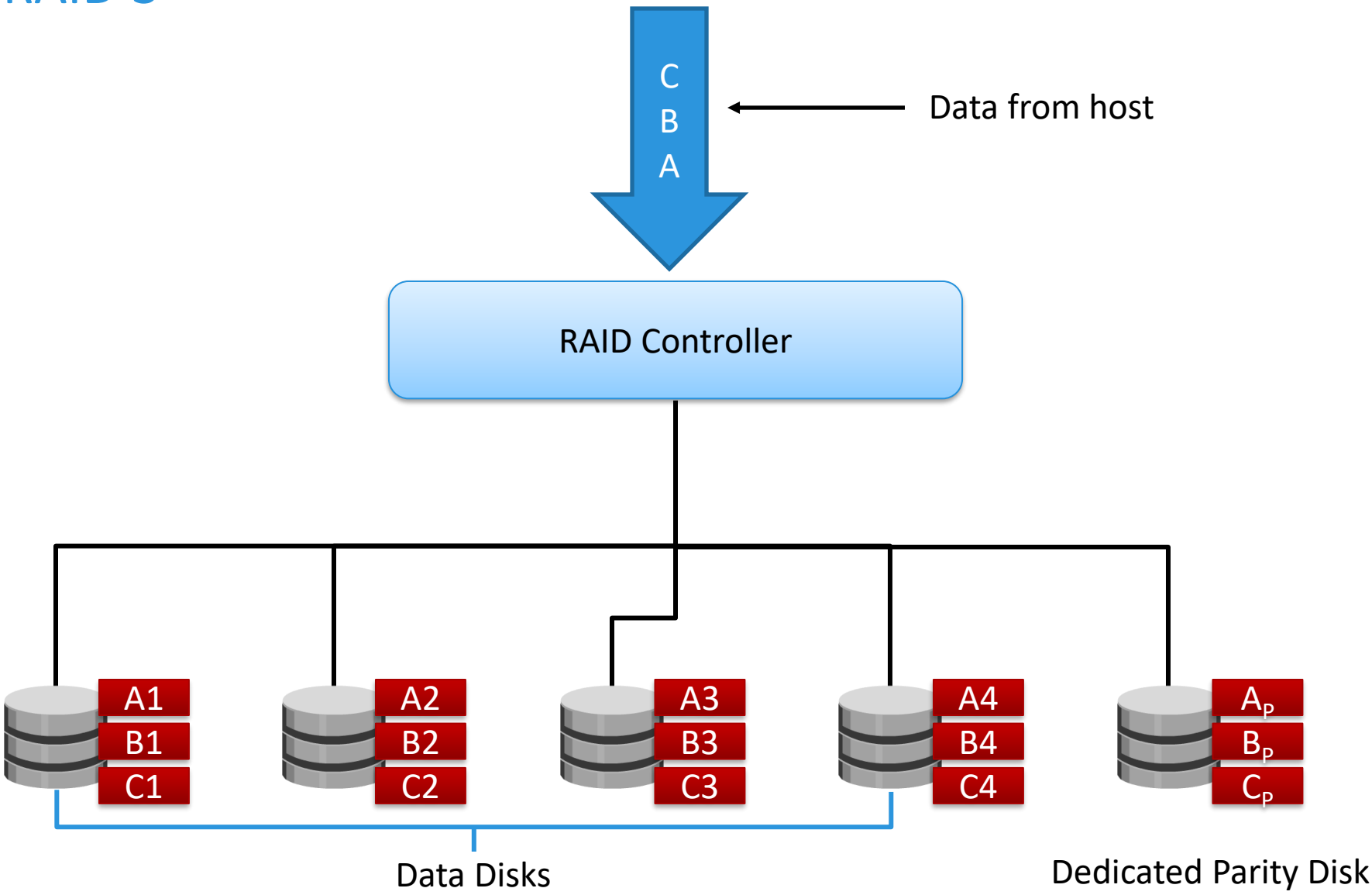
Input A	Input B	XOR Output
0	0	0
0	1	1
1	0	1
1	1	0

RAID 4

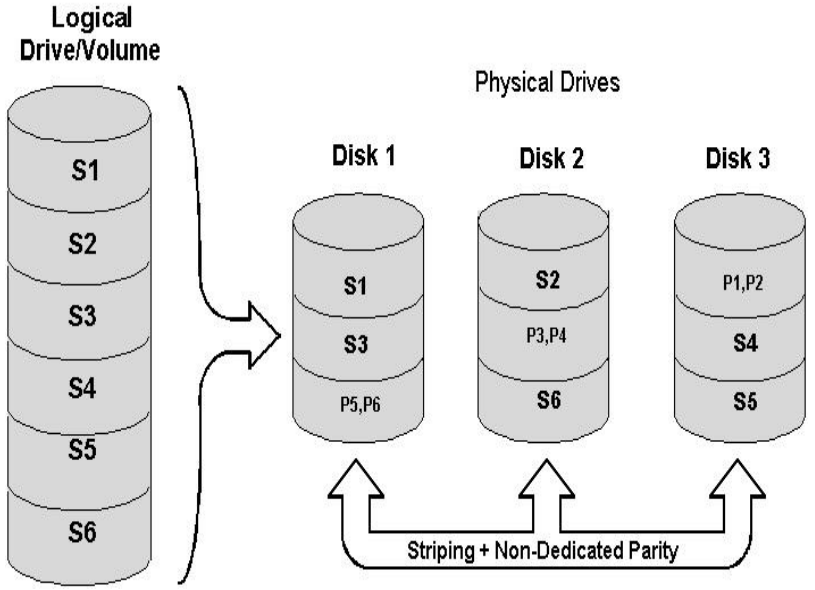
- very similar to RAID 3. RAID 4 uses **block level** striping with parity disk. (Block> Byte)
- Pros? advantage of block level change the stripe size to suit your application needs.



RAID 3


C
B
A
Data from host
RAID Controller
A1
B1
C1
A2
B2
C2
A3
B3
C3
A4
B4
C4
A_P
B_P
C_P
Data Disks
Dedicated Parity Disk

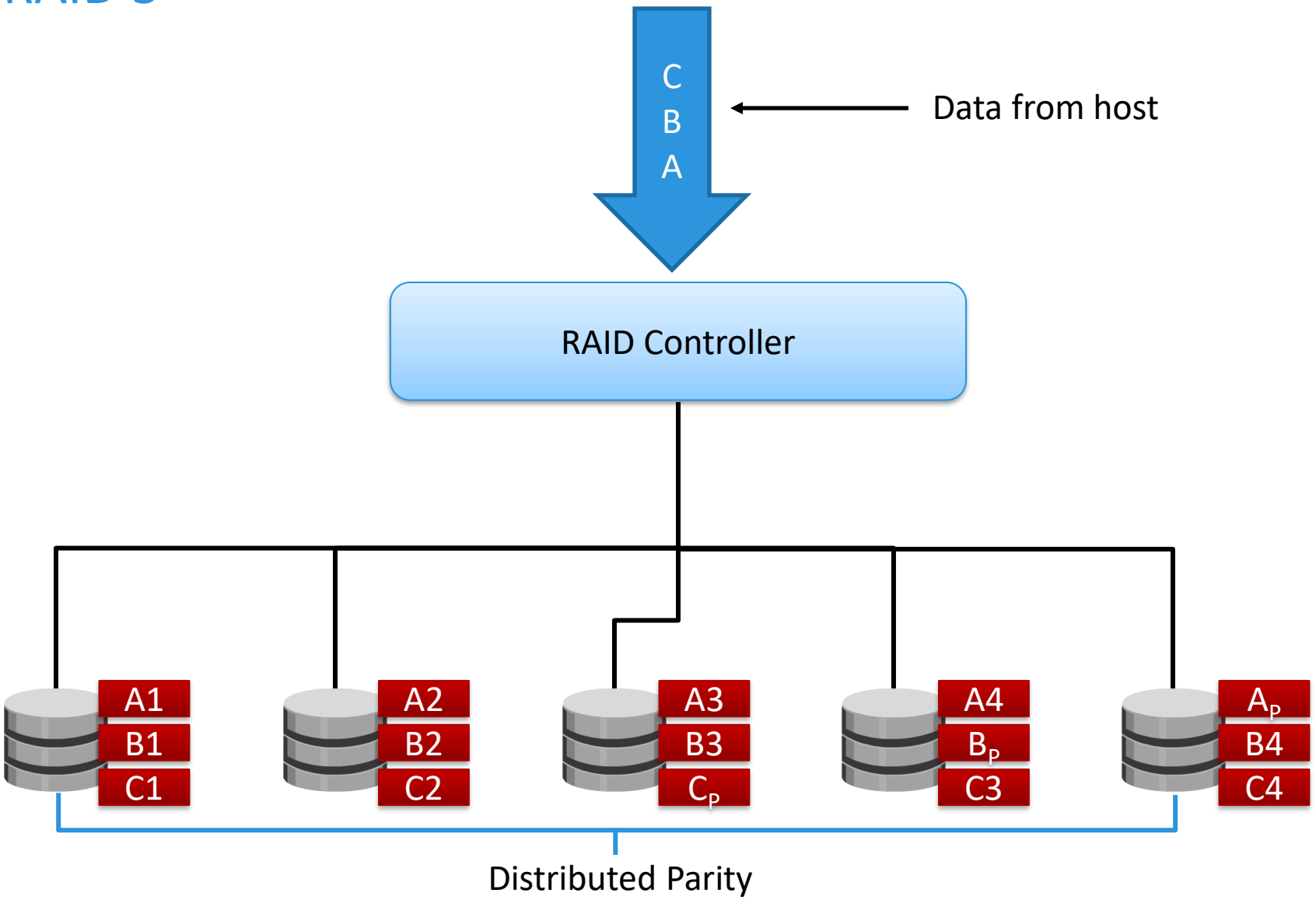
RAID 5



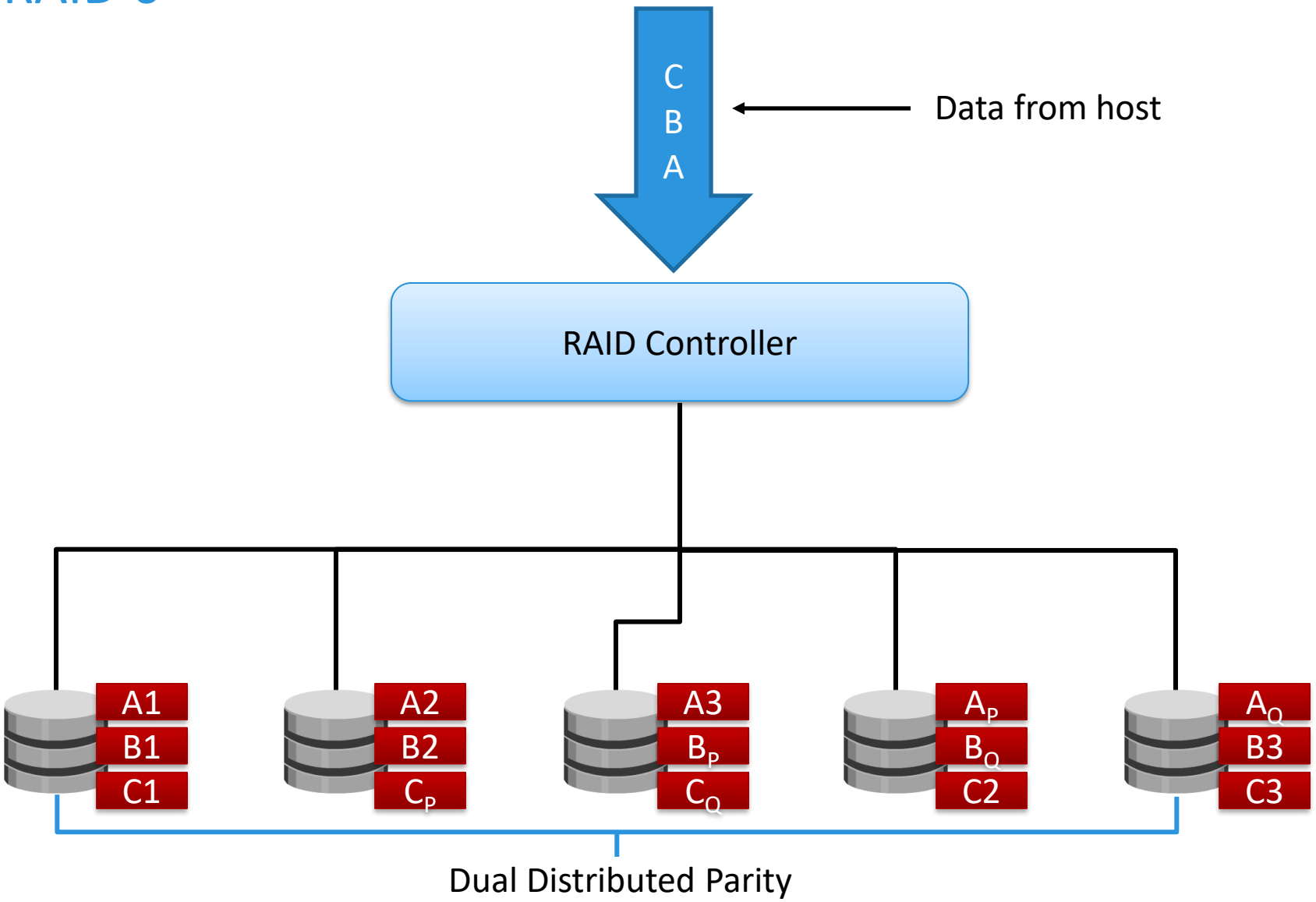
- Most popular member of the RAID family
  - combines block level striping

  with **distributed parity** for
    - good performance,
    - fault tolerance
    - storage efficiency.
- distributing parity stripes over a series of hard drives
  - Minimizes write bottlenecks of RAID levels 3 and 4
  - Provides relief to the concentration of write activity on a single drive, which in turn enhances overall system performance.
- RAID 5 is often used as an all-purpose RAID solution, but it is also used for database and file server applications.
- RAID 5 requires a minimum of three hard drives, but often costs less to implement than RAID 3 or 4.
- RAID recovery may be necessary if more than one disk fails.

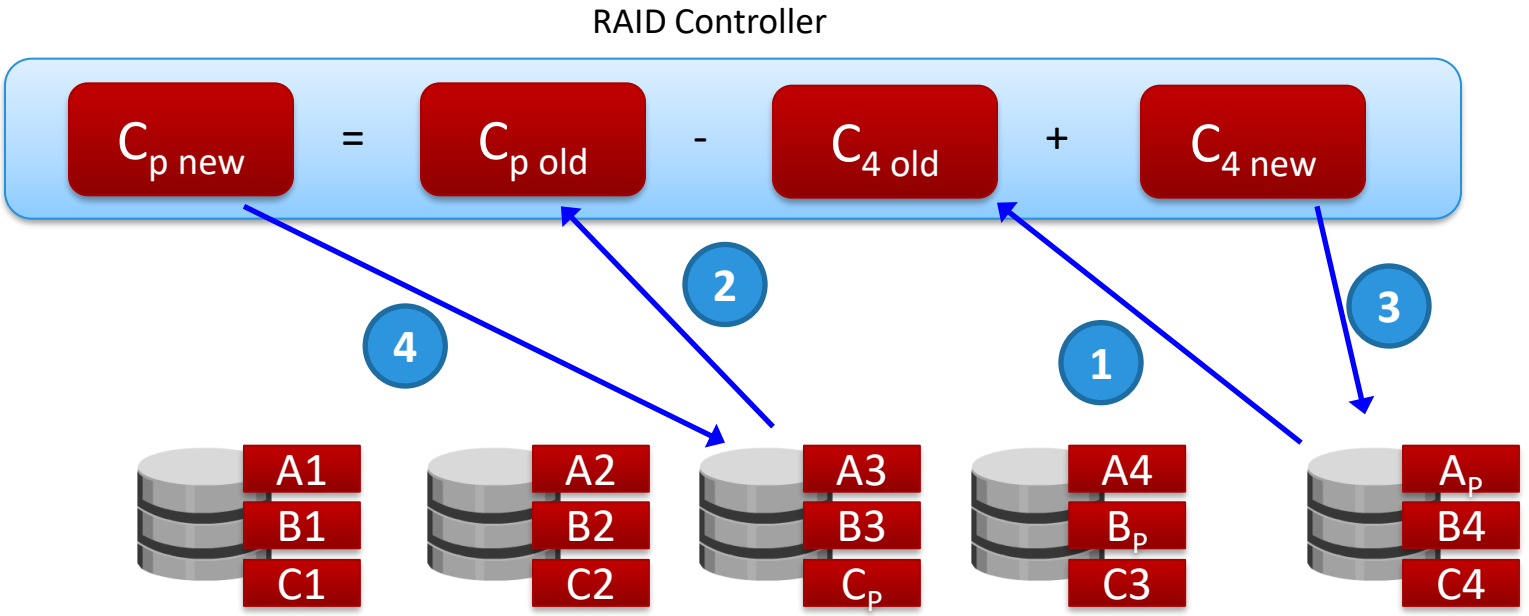
RAID 5


C
B
A
Data from host
RAID Controller
A1
B1
C1
A2
B2
C2
A3
B3
C_P
A4
B_P
C3
A_P
B4
C4
Distributed Parity

RAID 6


C
B
A
Data from host
RAID Controller
A1
B1
C1
A2
B2
C_P
A3
B_P
C_Q
A_P
B_Q
C2
A_Q
B3
C3
Dual Distributed Parity

RAID Impacts on Performance



- In RAID 5, every write (update) to a disk manifests as four I/O operations (2 disk reads and 2 disk writes)
- In RAID 6, every write (update) to a disk manifests as six I/O operations (3 disk reads and 3 disk writes)
- In RAID 1, every write manifests as two I/O operations (2 disk writes)

RAID Comparison

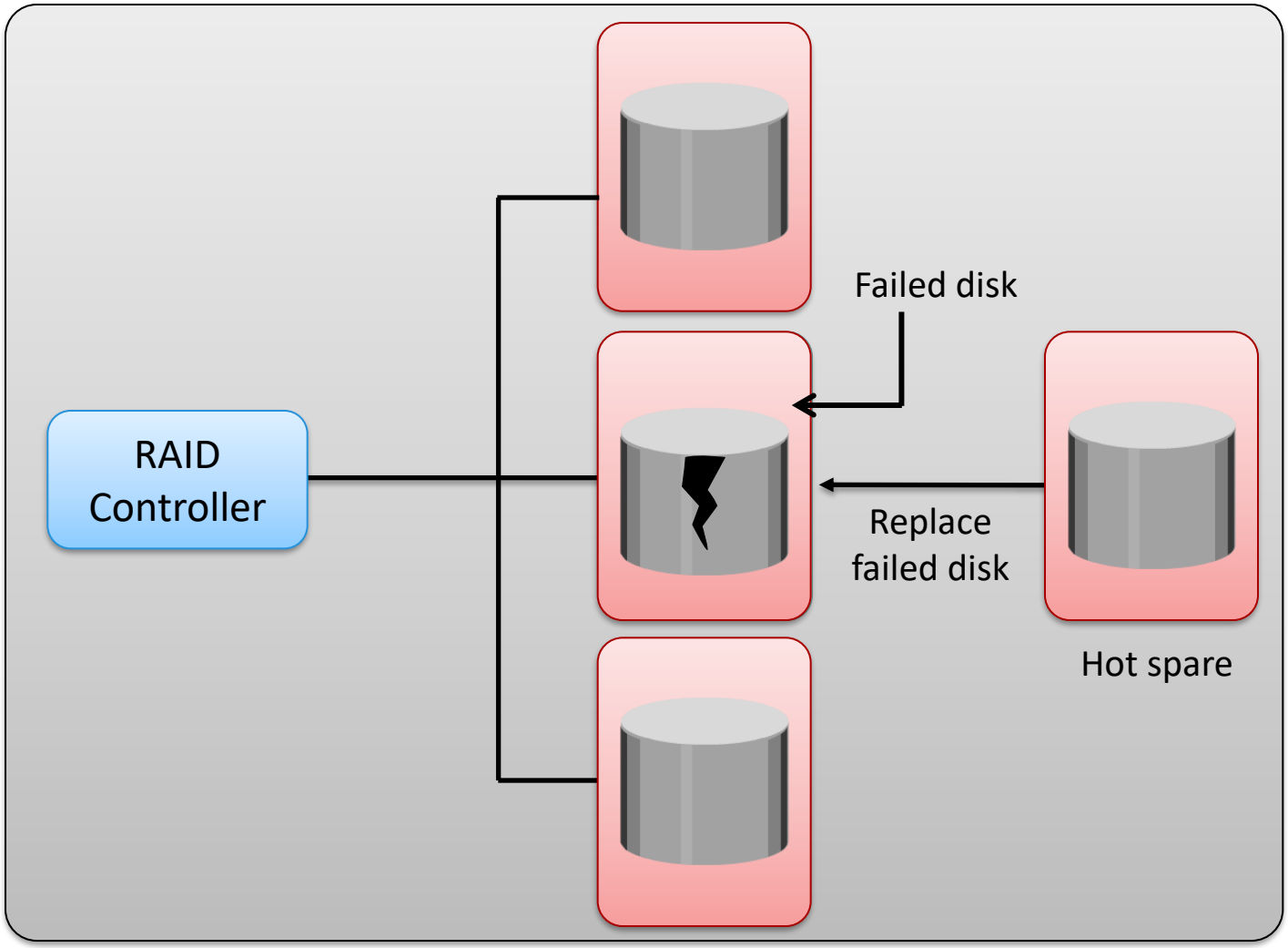
RAID level	Min disks	Available storage capacity (%)	Read performance	Write performance	Write penalty	Protection
1	2	50	Better than single disk	Slower than single disk, because every write must be committed to all disks	Moderate	Mirror
1+0	4	50	Good	Good	Moderate	Mirror
3	3	[(n-1)/n]*100	Fair for random reads and good for sequential reads	Poor to fair for small random writes fair for large, sequential writes	High	Parity (Supports single disk failure)
5	3	[(n-1)/n]*100	Good for random and sequential reads	Fair for random and sequential writes	High	Parity (Supports single disk failure)
6	4	[(n-2)/n]*100	Good for random and sequential reads	Poor to fair for random and sequential writes	Very High	Parity (Supports two disk failures)

where n = number of disks

Suitable RAID Levels for Different Applications

- RAID 1+0
  - Suitable for applications with small, random, and write intensive (writes typically greater than 30%) I/O profile
  - Example: OLTP, RDBMS – Temp space
- RAID 3
  - Large, sequential read and write
  - Example: data backup and multimedia streaming
- RAID 5 and 6
  - Small, random workload (writes typically less than 30%)
  - Example: email, RDBMS – Data entry

Hot Spare



Module 3: Summary

Key points covered in this module:

- RAID implementation methods and techniques
- Common RAID levels
- RAID write penalty
- Compare RAID levels based on their cost and performance